BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 026 · Wired and Wireless Networks

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What happens if the central switch of a star network fails?

Q2. More users share a limited network link. What may happen?
A. More bandwidth per user
B. Less bandwidth per user
C. Each device gains RAM
D. All delays disappear

Q3. What distinguishes a WAN?

Q4. Convert denary 150 into eight-bit unsigned binary.

Do Now Recall previous learning | **I Do** Read new knowledge | **We Do** Practise with guidance | **You Do** Work independently | **Review & Improve** Check my understanding

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

Wired connection — A link using a physical cable.

Copper cable — Transmission medium carrying electrical signals.

Fibre optic — Transmission medium carrying signals as light.

Wireless connection — Communication without a connecting cable, often radio.

Interference — Unwanted signals disrupting data transmission.

Encryption — Scrambling data so unintended recipients cannot read it without the key.

New learning

Copper cables: Wired copper connections carry electrical signals and are affordable, but have speed/distance limits.

Fibre-optic cables: Fibre carries pulses of light, supporting high rates and long distances; installation can be more costly.

Wireless connections: Wi-Fi uses radio waves so devices can move without cables, but connections may experience interference and interception.

Choose a connection: Use the scenario: high bandwidth, cost, mobility, length of cable, interference and security matter.

Worked example

A college connects two buildings a long distance apart. Explain why fibre optic may be more suitable than copper. [4 marks]

1. The connection is between buildings over a substantial distance.

2. Fibre uses light pulses [1] and is suitable for high-speed, longer-distance links [1].

3. Copper carries electrical signals [1] and has greater limitations on speed/distance for such a link [1].

4. Two contrasts, each explained in context, make four points.

We Do · Practise with guidance

Guided 1

A cafe provides wireless access to customers using phones and tablets. Explain two reasons Wi-Fi is suitable. [4 marks]

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Q3. Copper carries electrical signals [1] and is cost-effective for local wired links [1]. Fibre carries light [1] and offers high speed/long distance [1]. Wireless uses radio [1] and offers mobility but may suffer interference/interception [1].

Exit Ticket · Model answer

False

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed:

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. If the central switch fails, devices using it lose contact. A broken cable usually affects just one workstation.

Q2. B — Less bandwidth per user

Q3. A WAN links networks over a large area, often using links provided by other organisations.

Q4. (a) 10010110.

I Do / We Do pages · Guided practice

Q1. Wireless uses radio rather than cables to each device [1], allowing users to move while connected [1]. Devices can join without dedicated physical ports [1], making guest access more flexible [1].

Q2. Wireless signals can be intercepted by nearby devices [1], creating a confidentiality risk [1]. Encryption scrambles transmitted data [1], so captured content cannot be understood without the key [1].

Q3. Wired cables provide a stable link less affected by radio interference [1], improving reliability [1]. The cable restricts device movement [1], reducing flexibility if machines are relocated [1].

You Do pages · Code First

```
fileMb = 120
speedMbps = 60
print(fileMb DIV speedMbps)
```

Expected result: 2

You Do pages · Independent questions

Q1. Interference, interception/security risk or variable signal quality [1].

Q2. Fibre carries light-based signals [1] and is well suited to high-speed/long-distance links [1]. Installing fibre can be more costly or difficult [1], increasing the cost/complexity of the research station's connection [1].

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Worked method: Wireless uses radio rather than cables to each device [1], allowing users to move while connected [1]. Devices can join without dedicated physical ports [1], making guest access more flexible [1].

Guided 2

A business handles confidential data over Wi-Fi. Explain two risks and how encryption helps. [4 marks]

Worked method: Wireless signals can be intercepted by nearby devices [1], creating a confidentiality risk [1]. Encryption scrambles transmitted data [1], so captured content cannot be understood without the key [1].

Guided 3

A hospital must connect fixed machines reliably. Explain one benefit and one drawback of a wired connection. [4 marks]

Worked method: Wired cables provide a stable link less affected by radio interference [1], improving reliability [1]. The cable restricts device movement [1], reducing flexibility if machines are relocated [1].

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Code First · Predict → Write → Test → Improve

Predict the ideal transfer time in seconds.

```
fileMb = 120
speedMbps = 60
print(fileMb DIV speedMbps)
```

predict

Question 1

State one disadvantage of wireless transmission. [1 mark] Source: Adapted from your Paper 1 revision booklet, 1.3.2.

Question 2

A research station must connect an outdoor field laboratory to its main building. Explain one advantage and one limitation of using fibre-optic cable for this long link. [4 marks] Source: Original J277-style question, not an OCR past-paper original.

Question 3

Compare copper, fibre and wireless communication for a school. [6 marks] Source: Original J277-style question, not an OCR past-paper original.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Exit Ticket · Answer before checking

A wireless connection always needs a separate cable from every phone to the access point.